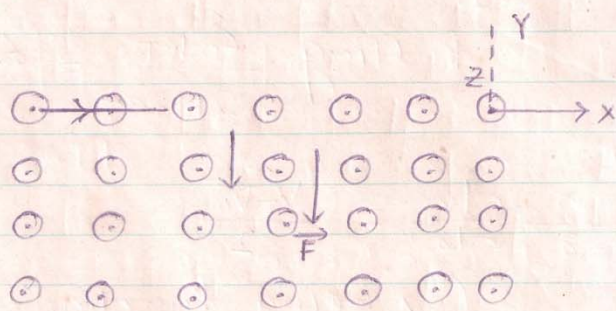




Plasma behaviour in a Magnetic field:

Case I Charged particles in a homogenous magnetic field in absence of any electric field
i.e. $\vec{E} = 0$



Given:

$B =$ The induction vector of the uniform magnetic field; chosen perpendicular to the plane of the

paper and directed towards the observer.

m and $q =$ mass & charge respectively of the particles that enters into this magnetic field with a velocity v , at right angles to the field say along x -axis as shown in the figure.

Force experienced by the moving charge in



a uniform magnetic field :

$$\vec{F} = q (\vec{v} \times \vec{B})$$

$$|\vec{F}| = q (vB \sin 90^\circ)$$

$$F = qvB \quad \text{--- (1)}$$

Applying the right hand rule for vector product; curl the finger of right hand from 1st vector $\vec{v}$ to 2nd vector $\vec{B}$ the thumb gives the product vector. we get the direction of force $\vec{F}$ along $-y$ axis in the fig shown. Thus the charged particle entering into the magnetic field at right angles to the field experiences a force of constant magnitude qvB and the direction perpendicular to the direction of velocity and therefore the particle describes a circular path of constant radius given by ;

$$F = qvB = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB} \quad \text{--- (1)} \quad \text{This is the radius of}$$

the orbit and is known as "LARMOR'S Radius".

One angular frequency (ie the angular velocity) ω_c is given by

$$\omega_c = \frac{v}{r} = \frac{v}{\frac{mv}{qB}}$$

$$\omega_c = \frac{qB}{m} \quad \text{--- (2)}$$

This frequency is known as



Cyclotron frequency.

We now assume that the particle enters into the same magnetic field at any angle θ other than 90° , with the direction of the field.

Let α be the angle made by the velocity vector with the direction of the induction vector $\vec{B}$.

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Let us resolve the velocity vector $\vec{v}$ into two mutually perpendicular components along & $\perp$ to the vector $\vec{B}$.

Component of $\vec{v}$ along $\vec{B}$ is $V_{\parallel} = V \cos \alpha$

Component of $\vec{v}$ along $\perp$ to $\vec{B}$ is $V_{\perp} = V \sin \alpha$

Force experienced by the charged particle due to its component of velocity $\parallel$ to the direction of magnetic field:

$$\vec{F}_1 = q (\vec{V}_{\parallel} \times \vec{B})$$

$$|\vec{F}_1| = q V_{\parallel} B \sin 0 = 0$$

Therefore the charged particle experience force only due to its component of velocity along $\perp$ to the direction of the field

$$\vec{F}_2 = q (\vec{V}_{\perp} \times \vec{B}) = q V_{\perp} B \sin 90^\circ$$

$|\vec{F}_2| = q v \sin \alpha B$ and therefore being perpendicular to the velocity vector $\vec{V}_{\perp}$, the

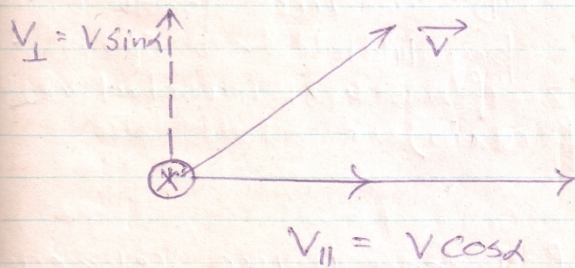


$$qVB \sin \alpha = \frac{mv^2 \sin^2 \alpha}{R} \quad \therefore \boxed{R = \frac{mv \sin \alpha}{qB}} \quad 55$$

particle describes a uniform circular motion of radius $R = \frac{mv \sin \alpha}{qB}$

Case II: Charged particle in crossed-electric and magnetic fields.

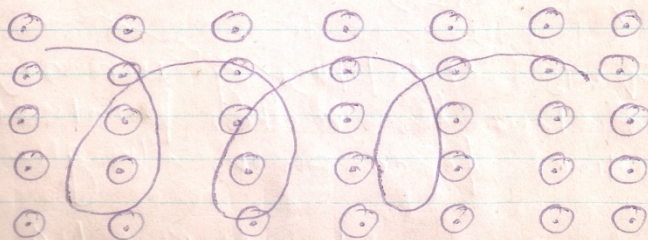
Suppose an additional homogenous electric field of intensity $\vec{E}$ is superimposed on the homogenous magnetic field of induction vector $\vec{B}$ at right angles to it as shown.



Thus the particle has one component of velocity $V \cos \alpha$ along the field direction which remains unchanged because there is no force

on the particle in that direction. The other component of velocity $V \sin \alpha$ also remains unchanged in magnitude but changes its direction, resulting in circular motion.

Therefore the resulting move under the influence of two motions will be a spiral as shown. i.e. moves along the field direction $\vec{B}$. The particle is therefore

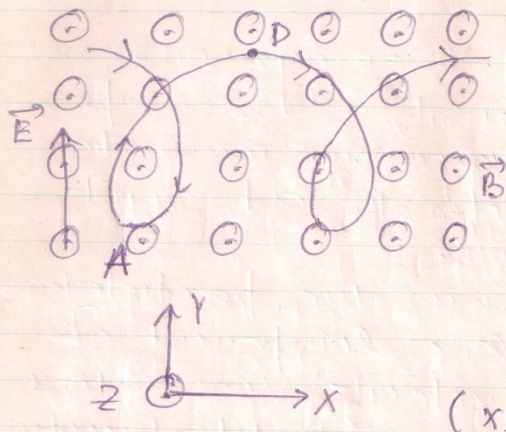


gyrating about the field direction & in a sense the particle



is locked to the field line.

Case II $\vec{B}$ is $\perp$ to the plane of the paper directed towards the observer (Z-axis), $\vec{E}$ is in the plane of the paper as shown (Y-axis)



Let us consider two special points A & D in the path of the particle as shown. (x, y, z) forms a R.H. handed system of axis

At the point A:

(i) The direction of force on the charged particle due to the electric field $\vec{F}_E = q\vec{E}$ along Y-axis.

(ii) The direction of velocity of the charged particle is along -X axis.

Force experienced due to the magnetic field:

$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

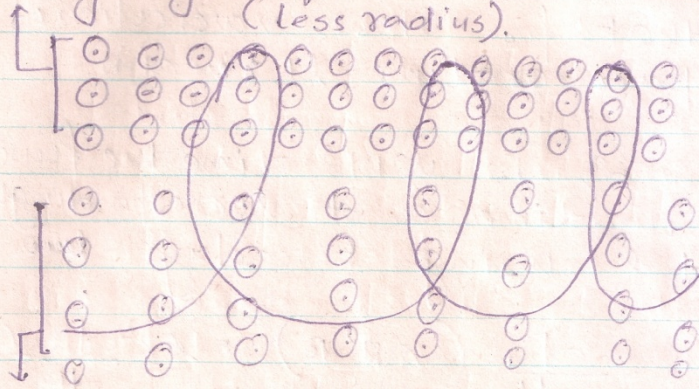
$F_m = qvB$ & applying the right hand rule for vector product direction of the force on is along +Y axis.

Resultant force on the particle at A

$$F = qE + qvB \text{ along Y-axis}$$



motion also involves a particle drift.
Strong magnetic field.
(less radius).



Weak magnetic field. (Large radius)

A gyrating charged particle possesses a magnetic moment μ which can be defined as

$\mu = \text{current} \times \text{area of the loop}$
 $= \text{charge} \times \text{no. of rotations per sec} \times \pi R^2$
 where $R = \text{Radius of the orbit}$.

$$\mu = q \cdot \frac{\omega}{2\pi} \cdot \pi R^2 \quad \left| \text{ But } \omega = \frac{V_{\perp}}{R} \right.$$

$$\therefore \mu = \frac{q V_{\perp} \cdot \pi R^2}{2\pi R} \quad \text{or} \quad \mu = \frac{q V_{\perp} R}{2} \quad \text{--- (A)}$$

We have seen that the radius of the orbit is given by $R = \frac{m V_{\perp}}{q B}$ --- (B)

and magnetic flux through the coil

$$\phi = B A = B \pi R^2 \quad \text{--- (C)}$$

Putting eqⁿ (C) in (A): $\mu = \frac{q}{2} \frac{(q B R) R}{m}$



Putting eqⁿ (D) : $\mu = \left(\frac{q^2}{2\pi m} \right) \phi$ — (E)

We now show that for slow variations of B , either in space or time μ remains constant.

Due to variation of B with time the flux also changes with time and correspondingly emf is induced & using Faraday's law

$$\mathcal{E} = \frac{1}{c} \frac{d\phi}{dt} = \frac{1}{c} \frac{d}{dt} (\pi R^2 B) = \pi R^2 \frac{dB}{dt} \quad \text{--- (F)}$$

When the change in the field is small, the work done upon a particle of charge q in one orbit is $[q\mathcal{E}]$ and this must be equal to the product of the rate of work done and the time taken by the particle to go through one orbit

i.e. $q\mathcal{E} = \frac{dW}{dt} \times T = \frac{dW}{dt} \cdot \frac{2\pi}{\omega} = \frac{dW}{dt} \cdot \frac{2\pi R}{v}$

$$q\mathcal{E} = \frac{2\pi R}{v} \frac{dW}{dB} \cdot \frac{dB}{dt} \quad \text{--- (G)}$$

Putting (F) in (G):

$$q \cdot \pi R^2 \frac{dB}{dt} = \frac{2\pi R}{v} \cdot \frac{dW}{dB} \cdot \frac{dB}{dt}$$

$$\frac{dW}{dB} = \frac{q R v}{2} \quad \text{--- (H)}$$

Putting eqⁿ (A) in (H): $\boxed{\frac{dW}{dB} = \mu} \quad \text{--- (I)}$

Now, $W = K.E = \frac{1}{2} m v^2$ Putting eqⁿ (C)

$$W = \frac{1}{2} m v^2 = \frac{1}{2} (mv) v = \frac{1}{2} (qBR) v \\ = \frac{1}{2} (q v R) B$$



Putting (A): $W = \mu B$ — (J)
Differentiating eqn (J) w.r to B

$$\frac{dW}{dB} = \mu + B \frac{d\mu}{dB} \quad \text{--- (K)}$$

Comparing eqn (I) + (K) we find that the 2nd term in eqn (K) must vanish i.e. $\frac{d\mu}{dB} = 0$.

Thus the magnetic moment remains constant inspite of the variation of B either in space or time.

The effect of such field variations therefore must be treated, taking care of constancy of μ . For μ to be constant the flux within particle orbit must remain constant. This is made possible if the radius of the orbit decreases in the region of higher B and vice-versa. Therefore we conclude that if a magnetic field that varies slowly in position as the magnetic lines of force converge the particle orbit is compressed in such a way to maintain Φ constant. The particle therefore is seen to move along the surface of a flux tube shown below.

